Design Integration Solutions

Given a use case that includes technical requirements, constraints, or drivers, specify the appropriate Salesforce application programming interface(s) (API) for the proposed solution.

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Introduction

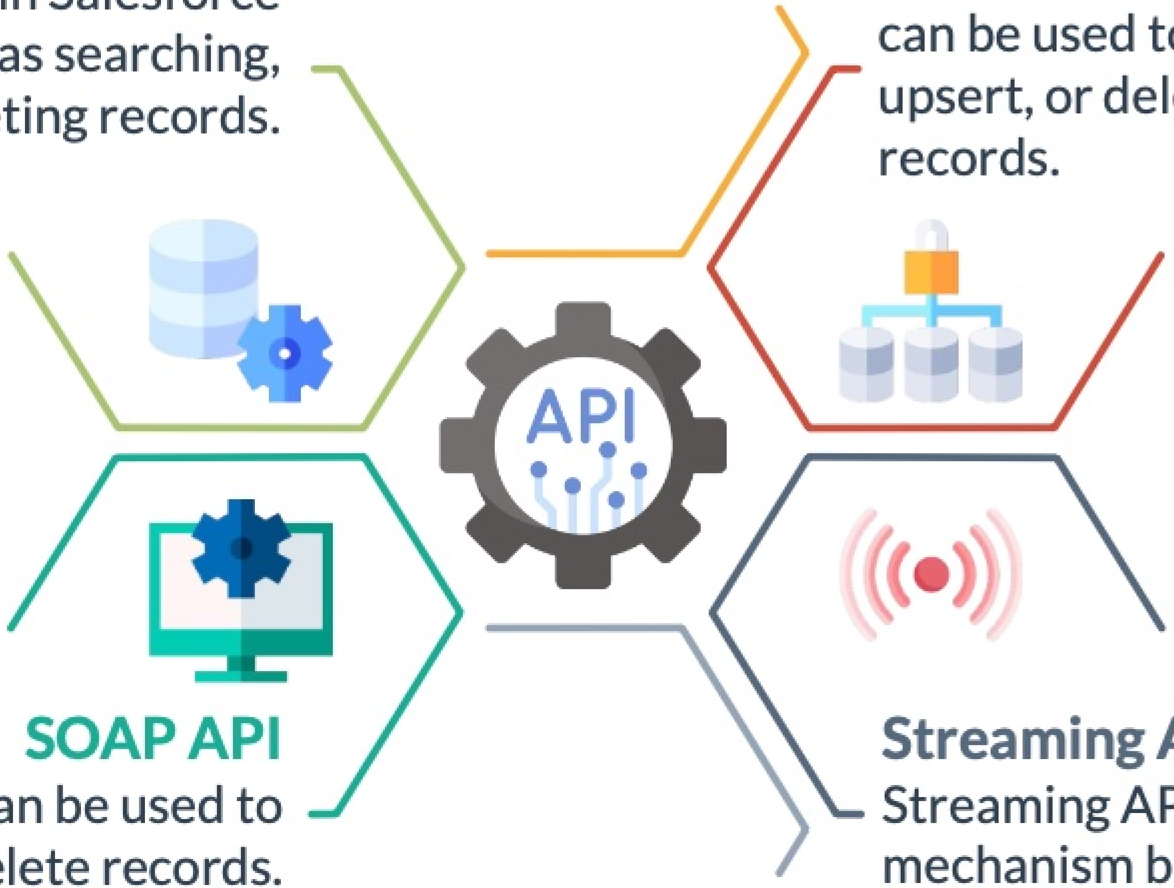
Salesforce provides access to various **APIs** that can be used for integration with external systems. **REST API** is a REST-based web services interface that can be used for programmatic access to data in Salesforce. Similarly, **SOAP API** can be used to retrieve, create, update, or delete Salesforce records.

Bulk API 2.0 is based on REST and is used to process large sets of data. It can be used to insert, update, upsert, or delete hundreds of thousands of records. **Streaming API** is a subscription-based mechanism based on CometD that enables real-time streaming of event messages.

Salesforce APIs

REST API
REST API can be used to implement programmatic access to data in Salesforce and perform operations such as searching, creating, updating, or deleting records.

Bulk API 2.0
Bulk API is based on REST principles and can be used to work with large data sets. It can be used to query, insert, update, upsert, or delete hundreds of thousands of records.



SOAP API
Like REST API, SOAP API can be used to retrieve, create, update, or delete records. SOAP API calls can be used to perform searches, convert leads, etc.

Streaming API
Streaming API is a subscription-based mechanism based on CometD that enables real-time streaming of event messages.

Salesforce APIs

Metadata API

Metadata API can be used to manage customizations in the org that is related to metadata such as custom object definitions, custom fields, page layouts, etc.

Tooling API

Tooling API can be used to build applications that can facilitate in the development and deployment of programmatic customizations in an org.

Connect REST API

Connect REST API can be used to provide programmatic access to specific resources in Salesforce and build a custom user interface for feeds, users, and groups related to Chatter.

Analytics REST API

Analytics REST API can be used to provide programmatic access to assets in CRM Analytics such as datasets, lenses, and dashboards.

Salesforce APIs

Salesforce offers various **APIs** that can be used for **integration with external systems**.

REST API

REST API provides a simple REST-based web services interface that can be used for programmatic access to data in Salesforce. A **resource** in REST API is identified by a named **URI (Uniform Resource Identifier)**, which can be accessed using standard HTTP methods. A resource can be used to **retrieve information, perform a search, create a record, update a record, or delete a record**. This API operates **synchronously** and communicates data in **JSON** or **XML** format.

BULK API 2.0

Bulk API 2.0 is based on **REST principles** and is optimized for working with **large data sets**. It can be used to **query, insert, update, upsert, or delete hundreds of thousands of records asynchronously**. This API operates **asynchronously** and processes data in **CSV** format.

Salesforce APIs

Salesforce offers various **APIs** that can be used for **integration with external systems**.

SOAP API

Like REST API, **SOAP API** can be used to **create, retrieve, update, or delete records**. It provides access to several calls that be used to perform searches, convert leads, etc. It can be used in any language that supports web services. This API operates **synchronously** and only supports the **XML** data format.

STREAMING API

Streaming API is a subscription-based mechanism based on **CometD** that enables **real-time streaming of event messages**. It allows receiving near real-time streams of data that are based on changes in Salesforce records or custom payloads. It utilizes **Bayeux** (a protocol that simulates push technology), operates **asynchronously**, and relays data in **JSON** format.

Salesforce APIs

Metadata API

To programmatically **retrieve**, **deploy**, **create**, **update**, and **delete** configurations in an org, the Metadata API can be used. A common use case is deploying changes from a related org to a production org. The **Salesforce Extensions for Visual Studio Code** and **Salesforce CLI** is built on top of the Metadata API. This API is **SOAP-based** and operates **asynchronously** and communicates data in **XML format** only.

Tooling API

The Tooling API can be used to allow external applications access to org metadata. In addition, it can be used to deploy **Apex**, **Visualforce**, and **Aura** components and provide development capabilities such as setting **checkpoints**, executing **anonymous Apex**, accessing **debug logs**, and much more. This API supports **REST** and **SOAP**, operates **synchronously**, and relay data using **JSON**, **XML**, or a **custom data format**.

Salesforce APIs

Connect REST API

To develop a **custom user interface** for **B2B Commerce**, **CMS managed content**, **Experience Cloud**, or **Chatter**, Connect REST API can be used. This API, for example, can be used to build a **mobile app**, integrate a **third-party web application** for notifying Chatter users, or display a **Chatter feed**. This API operates **synchronously** and communicates data in **JSON** or **XML** format. Photos, however, are processed **asynchronously.**

Analytics REST API

To programmatically access and manage resources such as **datasets**, **lenses**, and **dashboards** in **CRM Analytics** (formerly known as *Tableau* CRM), Analytics REST API can be used. This API operates **synchronously** and communicates using **JSON** or **XML** format.

Business Scenarios

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Business Scenario 1

Read the business scenario below and determine the most appropriate solutions for the company's requirements.

Cosmic Data Solutions is a company that offers data-backup and recovery solutions in the United States. It currently uses Salesforce and an ERP system to store and manage customer and employee data.

The sales department of the company uses Salesforce for managing customer accounts, contacts, and opportunities and a legacy CRM system for prospect management. Other departments, including the HR department, use the ERP system, which contains modules for various functional areas, such as recruiting, payroll, benefits, performance, accounting, inventory management, etc.

To reduce manual work and help the internal users become more productive, the CTO of the company has decided to revamp the systems. One of the main objectives is automating the primary business processes through integration. As a result, she has appointed a new Salesforce architect who needs to design solutions to some technical requirements using Salesforce APIs while considering any constraints or drivers.

Requirements & Solutions

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Requirement 1

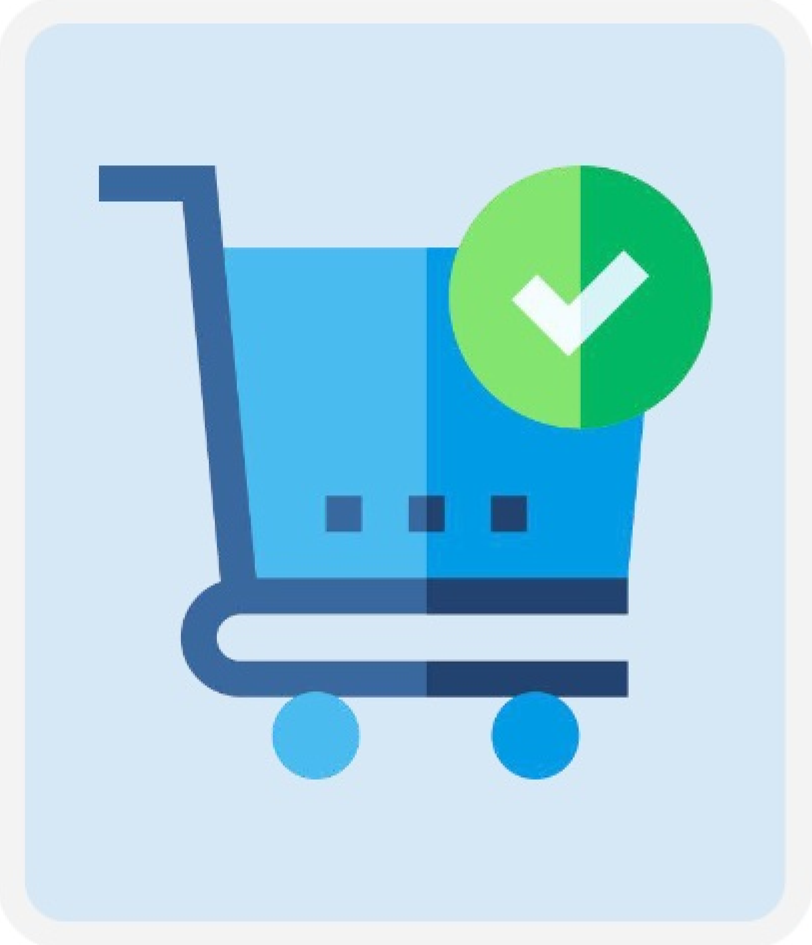
SCENARIO

The ERP system currently contains more than 2.5 million account records. Since the sales reps use only Salesforce for account management, the sales director would like to migrate the accounts from the ERP system to Salesforce. The system administrator of the company is only familiar with the use of the Data Import Wizard for loading records. However, since it cannot process more than 50,000 records, the Salesforce architect needs to recommend an alternative solution that provides high data throughput and ensures efficient loading of the large amount of data. One should be able to monitor the progress of the data load and reprocess any records that were not loaded successfully.

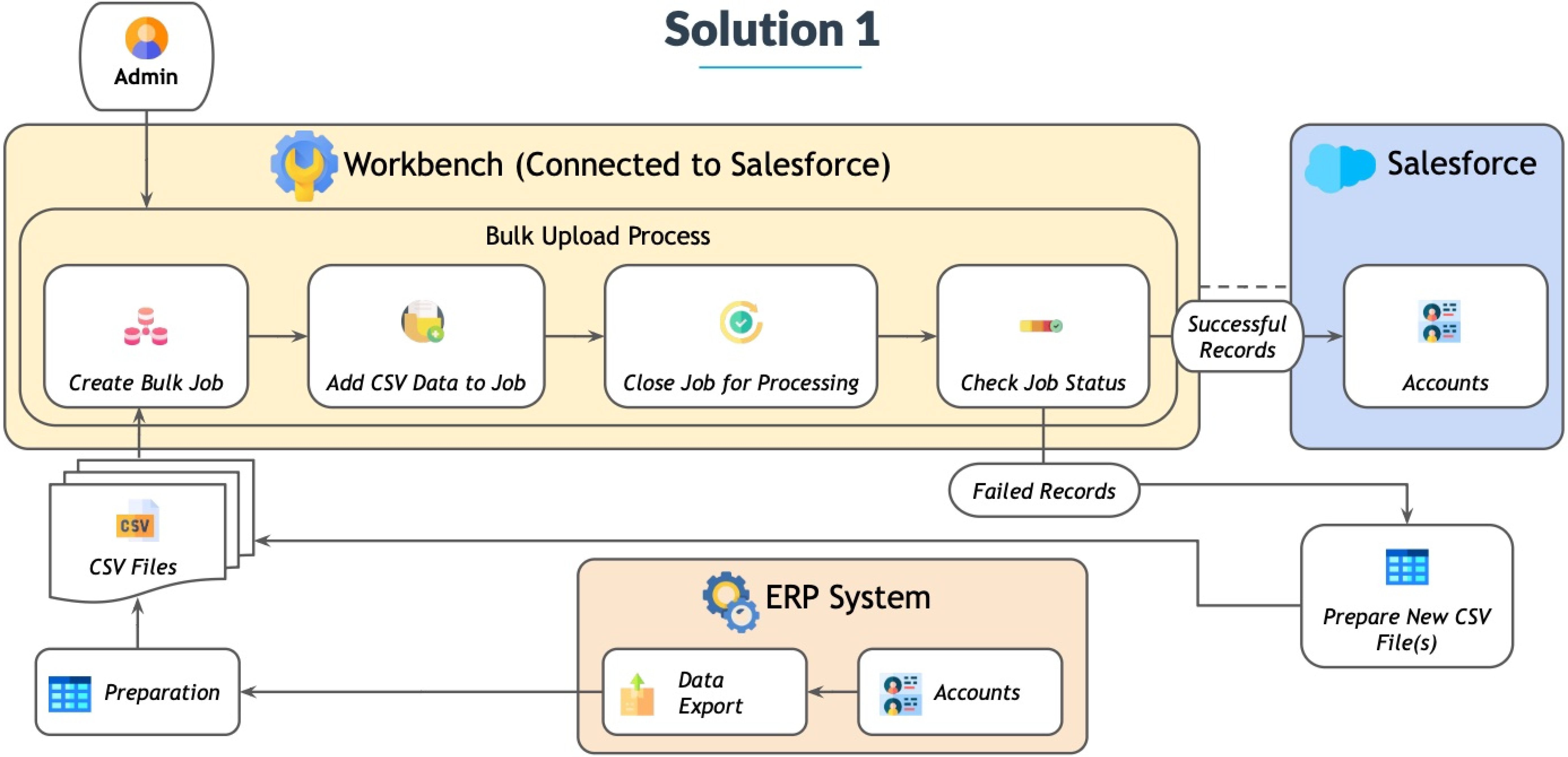
Solution 1

SOLUTION

Given the large number of records to be imported, **Bulk API 2.0** can be leveraged for the data load. The account records in the ERP system can be exported as **CSV files** and prepared for the data import. The **REST Explorer** in **Workbench** can be used to create a Bulk API job by submitting a **POST** request to the **/jobs/ingest** resource. The **job ID** in the response should be noted. Account data in a CSV file can be added to the job by submitting a **PUT** request to the **/jobs/ingest/ jobID/batches** resource. To process the account data, the job should be closed by submitting a **PATCH** request. The status of the job can be checked using Workbench or on the **Bulk Data Load Jobs** page in Salesforce Setup. Workbench can be used to obtain the job results in the form of successfully and unsuccessfully loaded records.



Solution 1



Admin
Workbench (Connected to Salesforce)
Bulk Upload Process
Create Bulk Job
Add CSV Data to Job
Close Job for Processing
Check Job Status
Successful Records
Salesforce
Accounts
Failed Records
Prepare New CSV File(s)
CSV
CSV Files
ERP System
Data Export
Accounts
Preparation

Requirement 2

SCENARIO

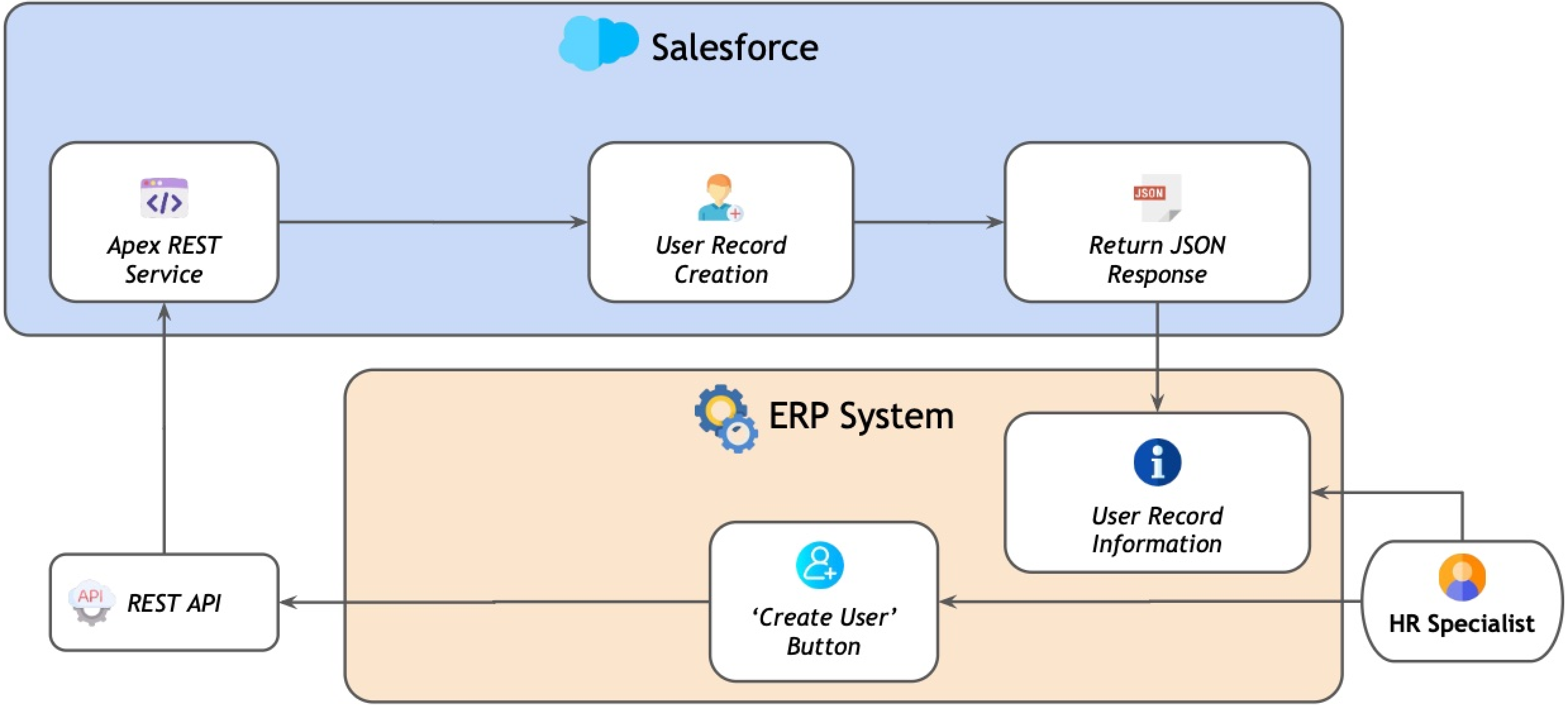
The HR department currently uses the ERP system for starting the recruiting process and managing candidates. Once a candidate reaches the ‘Recruited’ stage and requires access to Salesforce, an HR specialist should be able to create a user record from the ERP system using a button. When the button is clicked and the user record is created successfully in Salesforce, the HR specialist must receive a notification from Salesforce immediately. The solution architect would like to build a lightweight solution for this requirement and avoid having to work with (Web Services Description Language) WSDL or XML files.

Solution 2

SOLUTION

JSON is one of the two data formats supported by **REST** and is more lightweight than **XML**. SOAP only supports XML. In this case, the standard **REST API** can be used to perform the basic user creation. However, if custom logic such as a custom response or complex validation is required in the transaction, **Apex REST service** can be used instead. The Apex class used in a REST service is annotated with **@RestResource(urlMapping=*'/customUrl'*)**. To expose the Apex method that inserts the user record, it should be annotated with **@HttpPost**. It is important to note that REST API is **synchronous**, so an HR specialist who creates a user record from the ERP system immediately receives a response from Salesforce.

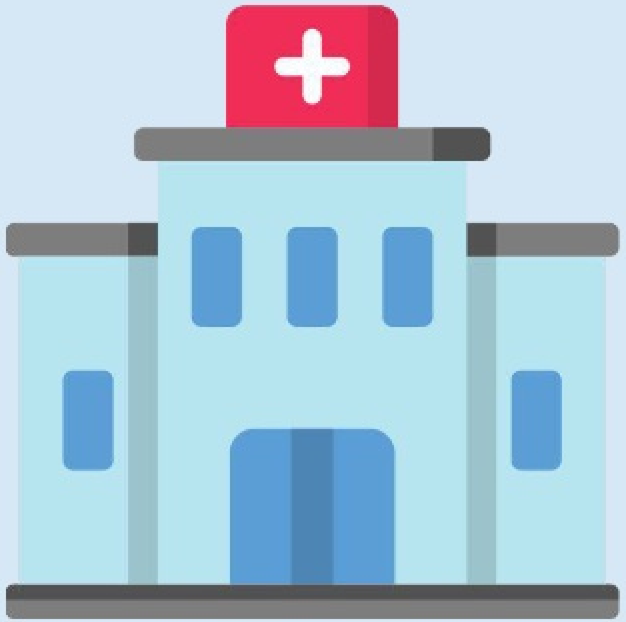
Solution 2



Salesforce
Apex REST Service
User Record Creation
Return JSON Response
ERP System
User Record Information
'Create User' Button
REST API
HR Specialist

Business Scenario 2

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Pharmaceuticals is a publicly-traded company that has heavily invested in Salesforce as an application platform. It has offices and warehouses in several countries, such as the United States, Australia, Germany, and France. Most of the company's customers are individuals who can purchase approved drugs directly from the company's website or through a regional business manager.

It uses Salesforce to capture account information when a patient places an order and a legacy CRM for prospect management. Order processing occurs in SAP, which currently does not exchange any data with Salesforce. The CTO of the company would like to improve the efficiency of the internal processes by integrating systems with one another. Therefore, a Salesforce architect has been hired whose main task is to recommend solutions to various technical requirements by analyzing the given constraints or drivers.

Requirements & Solutions

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Requirement 3

SCENARIO

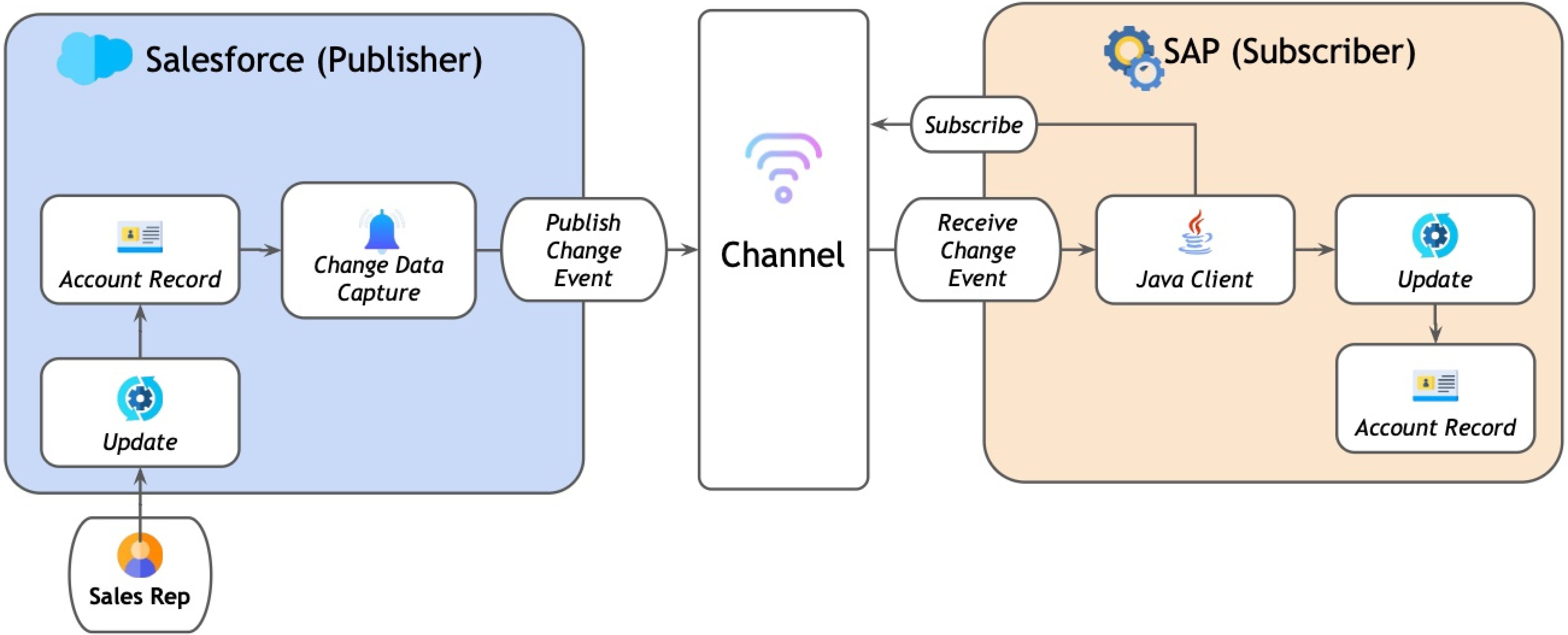
The regional business managers often need to update the account information of patients in Salesforce. When a patient's account record is updated in Salesforce, the corresponding patient information in SAP also needs to be updated manually. The sales director would like to automate this process to ensure that the business managers focus more on order management than data entry. The key fields that often get updated are Phone, Email, and Mailing Address. When any of these fields are updated on an account record in Salesforce, the change should be reflected in SAP as soon as possible so that the latest information is available in the system for order processing. Due to some limitations, a REST or SOAP API endpoint cannot be exposed in SAP for this use case.

Solution 3

SOLUTION

Since the update should be reflected in SAP in near real-time and the use of REST/SOAP callouts is not supported, **Streaming API**, which relies on the publish/subscribe pattern, can be utilized. **Change Data Capture**, a streaming API tool, can be configured to publish **any changes** made specifically to Account records as **change events**. In SAP, a **Java** client that uses the **EMP Connector** can be built and used to connect to **CometD** and **subscribe** to the **ChangeEvents** channel. Subscribing via the **Pub/Sub API** is also supported. When SAP receives a change event, it can automatically update the corresponding fields on the patient's account record in SAP.

Solution 3



Salesforce (Publisher)
Account Record
Change Data Capture
Update
Sales Rep
Publish Change Event
Channel
Subscribe
Receive Change Event
SAP (Subscriber)
Java Client
Update
Account Record

Requirement 4

SCENARIO

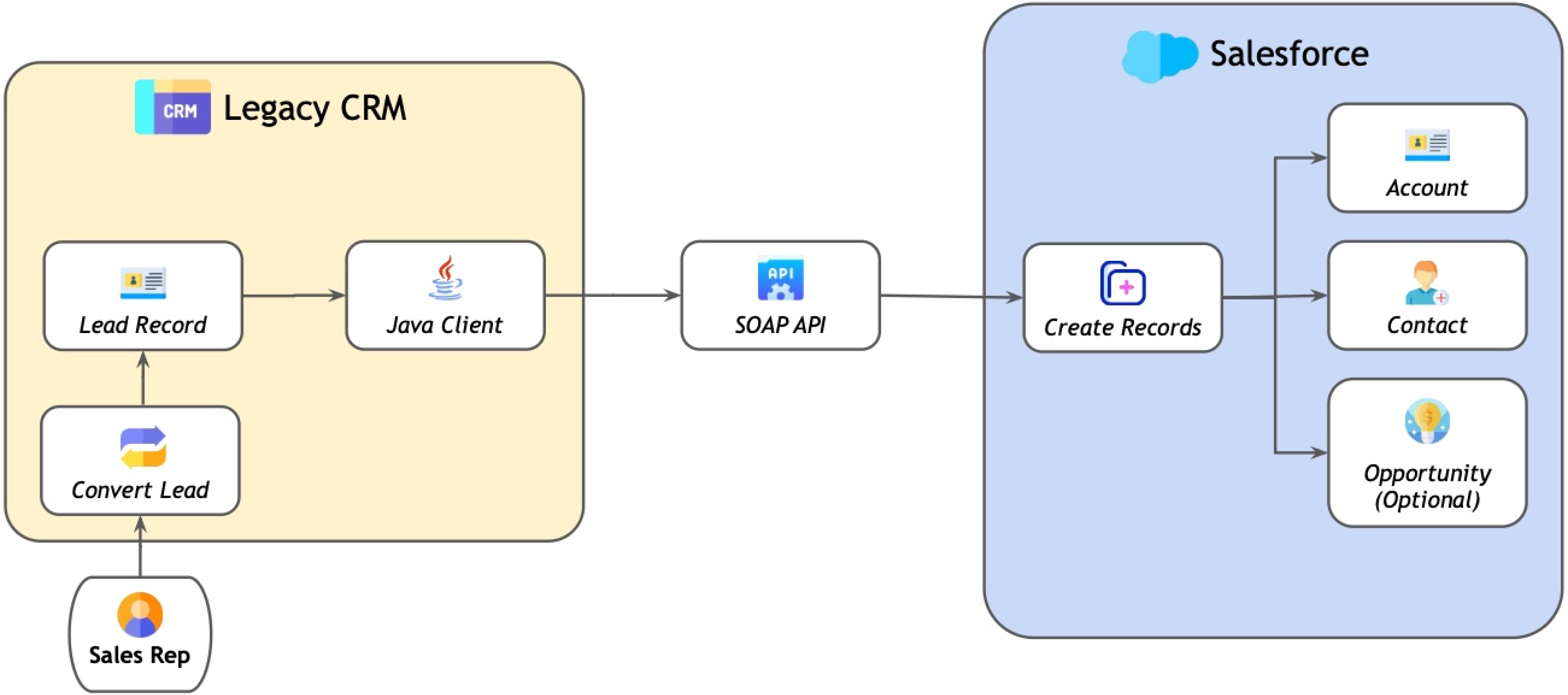
The sales director of the company wants to allow sales reps to quickly convert leads into accounts, contacts, and optionally opportunities in Salesforce from a user interface only they can access in the legacy CRM system. However, the Salesforce developer of the company is currently unavailable for programmatic development in Salesforce, so the lead conversion should not require writing Apex code. But a Java developer is available for writing code in the legacy CRM system for this requirement. The solution architect needs to recommend a solution while taking this into account.

Solution 4

SOLUTION

SOAP API can be used to meet this requirement. The **convertLead()** call provided by the API can be used to convert a lead into an account, contact, and (optionally) an opportunity. It is also possible to convert a lead into an account and a person account instead of a contact. Java code can be written for the client application in the legacy CRM system to connect to Salesforce via SOAP API and use the convertLead() call for lead conversion. The code can be associated with a button in a user interface that allows sales reps to convert leads.

Solution 4



Legacy CRM
CRM
Lead Record
Java Client
Convert Lead
Sales Rep
SOAP API
API
Salesforce
Create Records
Account
Contact
Opportunity
(Optional)

Learn More

- [Which API Do I Use?]
- [Which Salesforce API to Use When]
- [Get to Know the Salesforce Platform APIs]